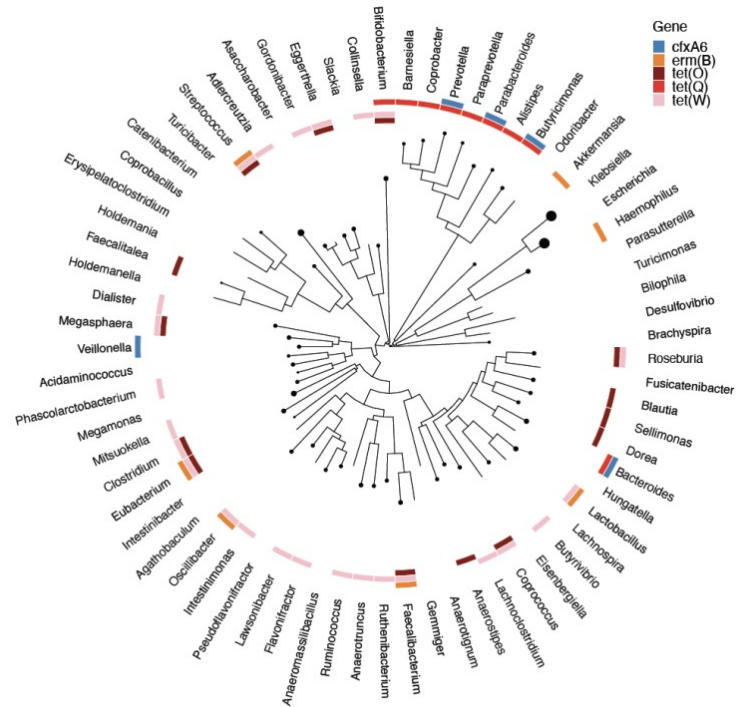


Statistical programming framework for multi-source data integration

Leo Lahti

datascience.utu.fi | @antagomir

Department of Computing,
University of Turku, Finland



Microbiome population cohorts: FINRISK

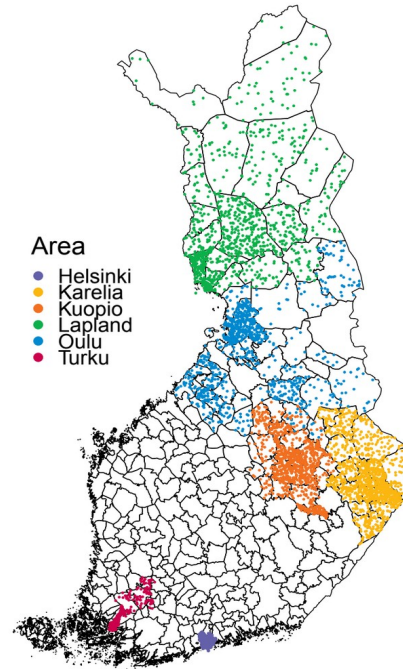
Prospective setup, data & methods

Long follow-up (since 2002)

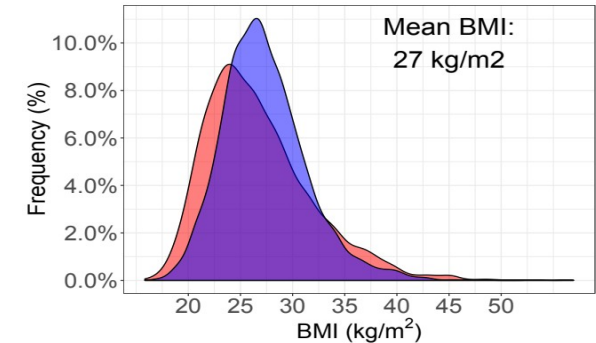
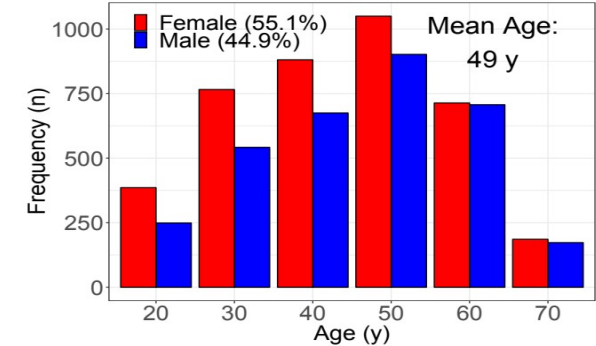
High quality register data

2002 ~7200+ adult stool samples:
omics and clinical measurements

Shallow shotgun metagenome
(~1M reads/sample) & 16S

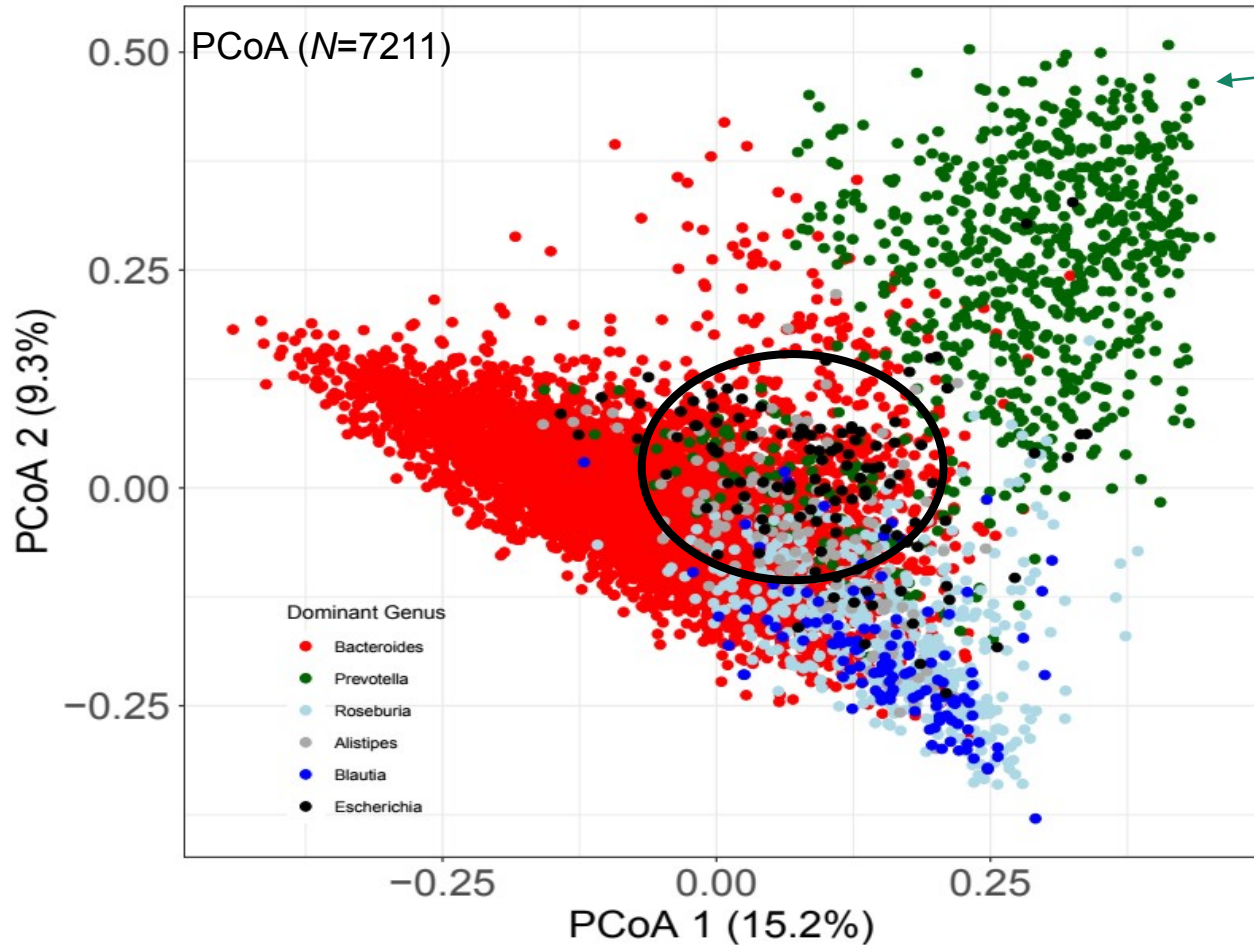


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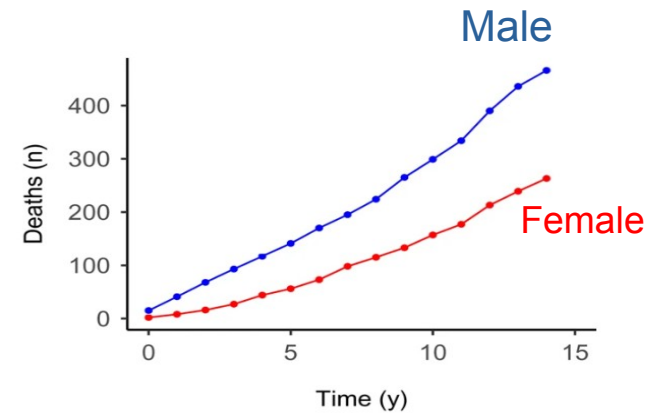
Salosensaari *et al.* Nat. Comm. 12:2671, 2021

Gut microbiome & mortality risk



You?

Increased mortality risk in Enterobacteriaceae-rich regions of this population landscape



Taxonomic signatures of cause-specific mortality risk in human gut microbiome

Aaro Salonen, Ville Lahti, Aki S. Havulinna, Guillaume Meric, Susan Chen, Markus Perola, Lisa Valsta, Georg Alifan, Michael Inou, Jerome D. Watrous, Teo Long, Rodolfo A. Salido, Karina Sanders, Caitriona Brennan, Gregory C. Humphrey, Jon G. Sanders, Mohit Jain, Pekka Jousilahti, Veikko Salonen, Rob Knight, Leo Lahti & Teemu Niiranen

Nature Communications 12, Article number: 2671 (2021) | [Cite this article](#)

11k Accesses | 16 Citations | 359 Altmetric | [Metrics](#)

Hologenomics & data layers

Microbiome

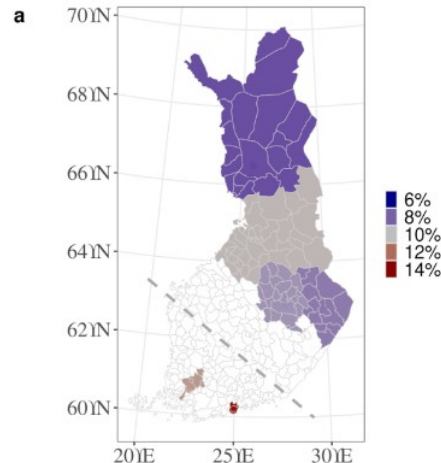
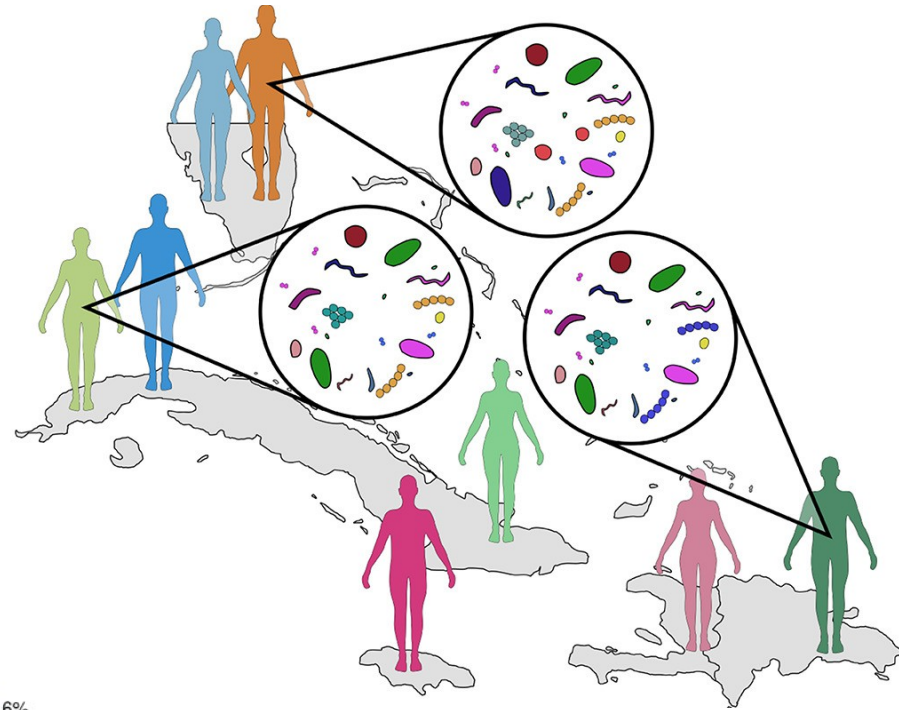
- gut metagenomes & 16S rRNA
- pathway / function predictions
- resistome profiles

Host

- Genotyping
- Exomes
- Circulating metabolites
- Clinical data

Environment

- e.g. population registers
- urban / rural



Minireview | [Open Access](#) | [CC BY](#)

Modelling spatial patterns in host-associated microbial communities

Matti O. Ruuskanen [✉](#) Guilhem Sommeria-Klein, Aki S. Havulinna, Teemu J. Niiranen, Leo Lahti

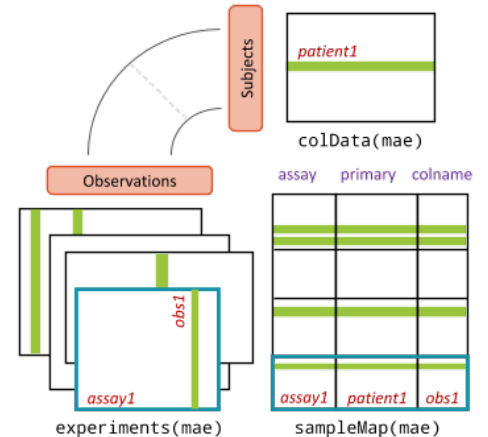
First published: 18 March 2021 | <https://doi.org/10.1111/1462-2920.15462> | Citations: 14

Hologenomics & data layers

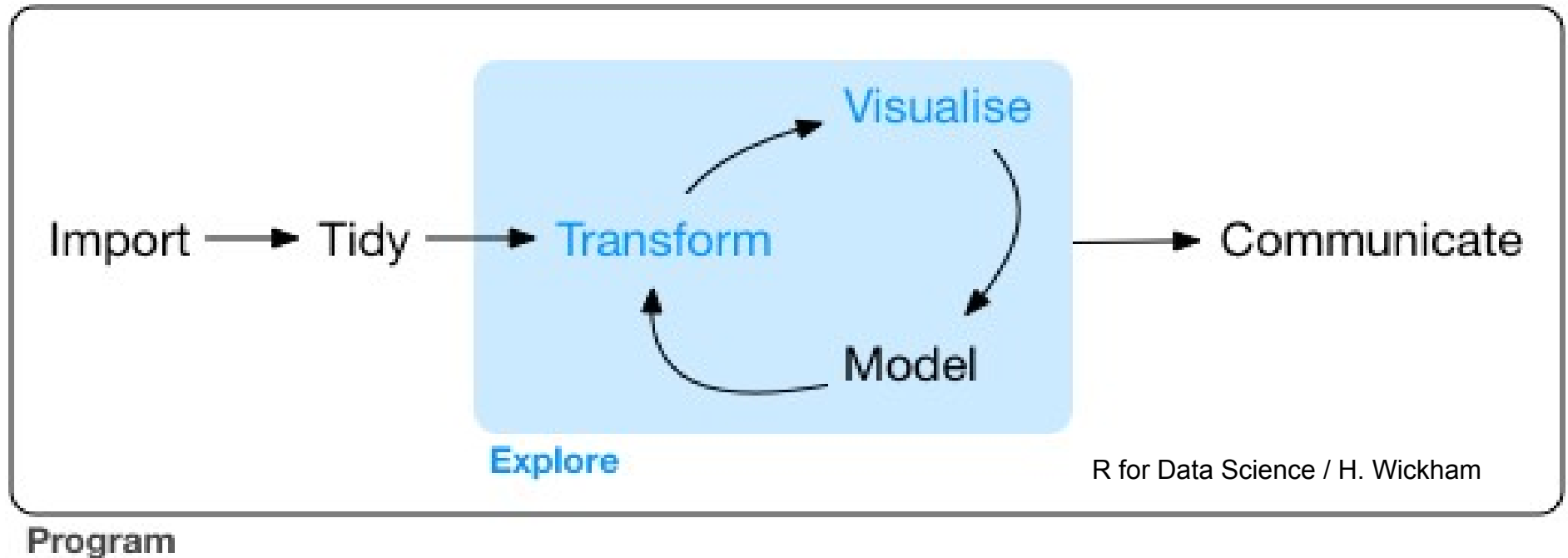
- Increasing sample sizes, depth & untargeted assays
- Spatio-temporal variation & resolution
- Multi-pipeline (processing, pipelines, transformations)
- Multi-level (hierarchies, phylogenies, nested designs)
- Multi-assay (e.g. 16S + metagenome)
- Multi-omic (taxonomic, functional)
- Multi-kingdom (bacteriome, virome, mykobiome, eukaryome)
- Multi-cohort (meta-analyses)

→ How to organize heterogeneous data sources?

→ How to build interoperable methods?



Data science workflow

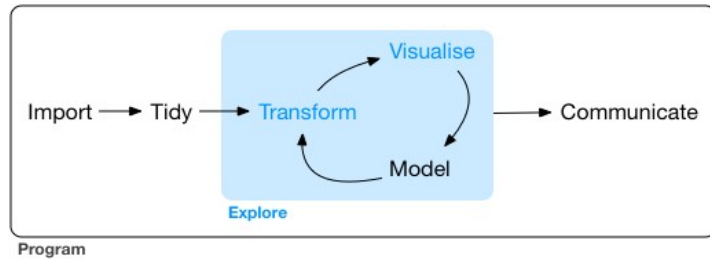


- Transparent
- Reproducible
- Robust

“What marks out modern science is not the conduct of experiments (alchemists conducted plenty of experiments), but the formation of a *critical community capable of assessing discoveries and replicating results.*” The Invention of Science: A New History of the Scientific Revolution, by David Wootton

Alchemy & algorithms: perspectives on the philosophy and history of open science

▼ [Leo Lahti](#), [Filipe da Silva](#), [Markus Petteri Laine](#), [Viivi Lähteenoja](#), [Mikko Tolonen](#)



A family of alchemists at work, an engraving by Philip Galle, after a painting by Pieter Bruegel the Elder, published by Hieronymus Cock, c.1558.

World's largest bioinformatics project:

- 2300+ packages
- Tens of thousands of users & publications
- Multi-omics integration

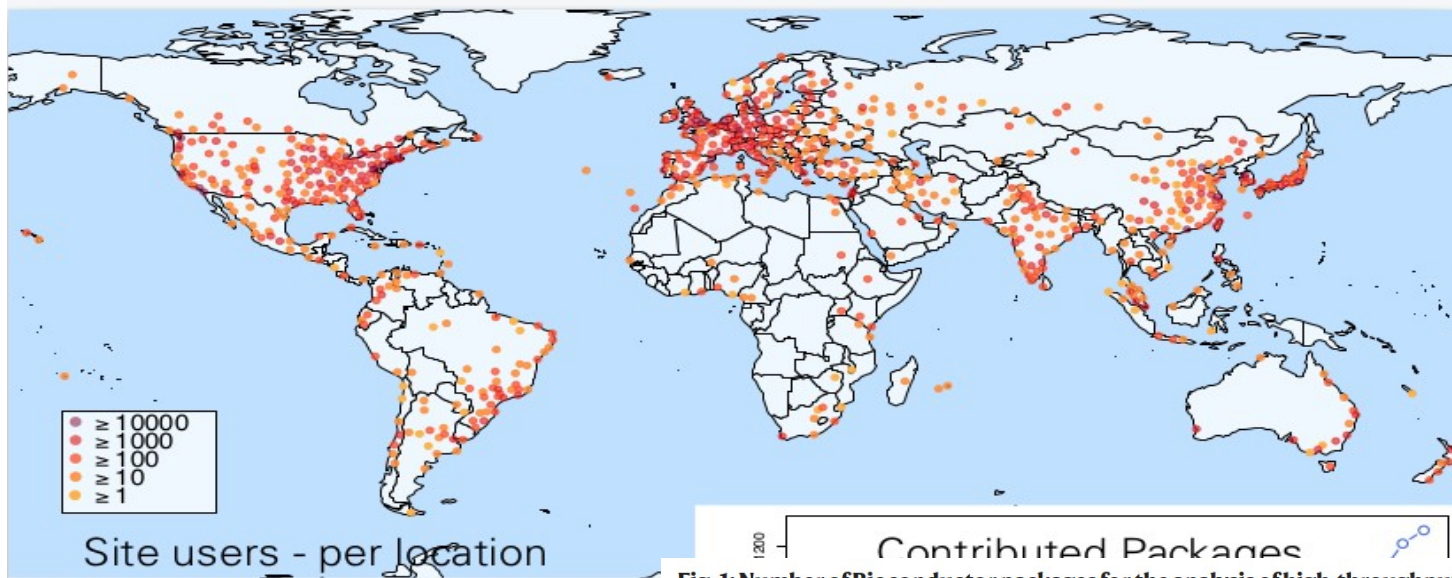
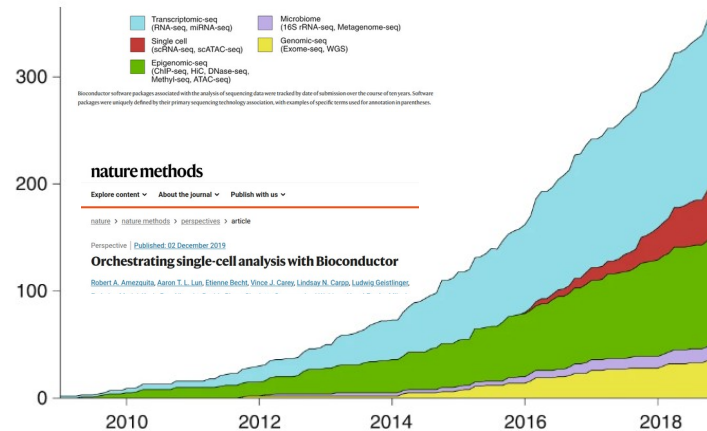


Fig. 1: Number of Bioconductor packages for the analysis of high-throughput sequencing data over ten years.

From: [Orchestrating single-cell analysis with Bioconductor](#)

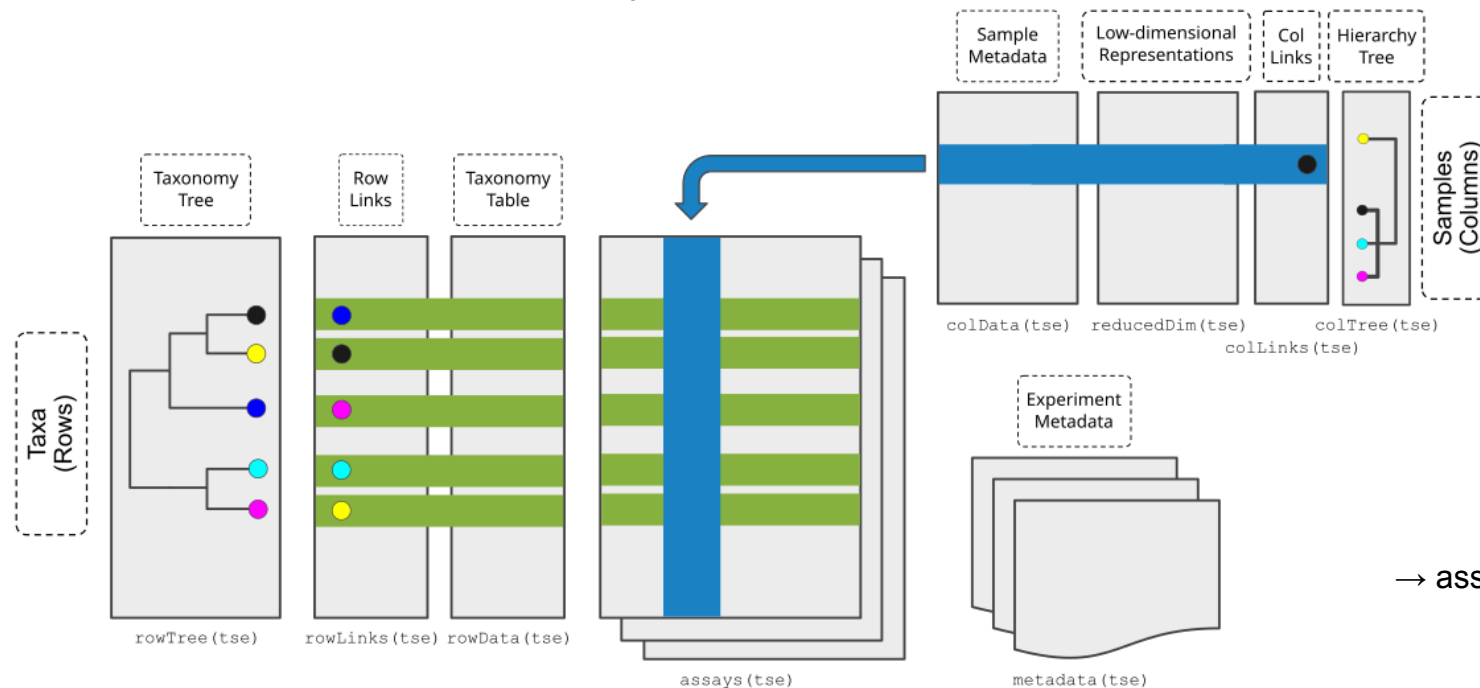
Number of R/Bioconductor packages for the analysis of sequencing data



Data containers: systematic multi-table integration

Commonly adopted data structures facilitate interoperability:

- (widely) shared methods & workflows
- more efficient & standardized analyses



Multiple assays
→ assays, altExp, reducedDim

Rich side information
→ sequences, metadata, rowData, colData

Hierarchies on sample & feature spaces
→ rowTree, colTree

FOCUS ON COMPUTER RESOURCES | NOVEMBER 01 2017

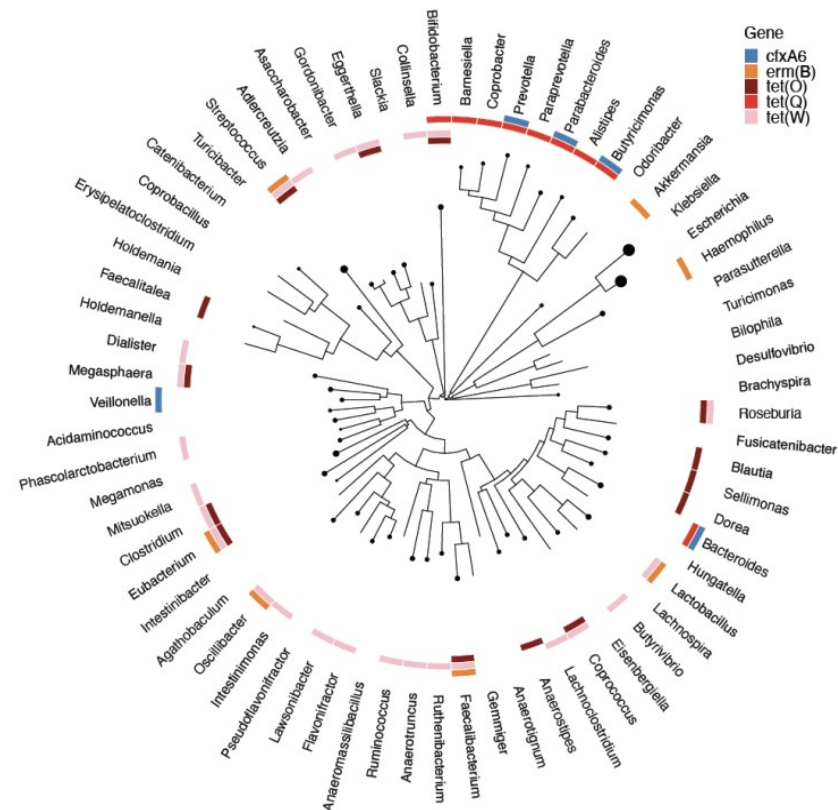
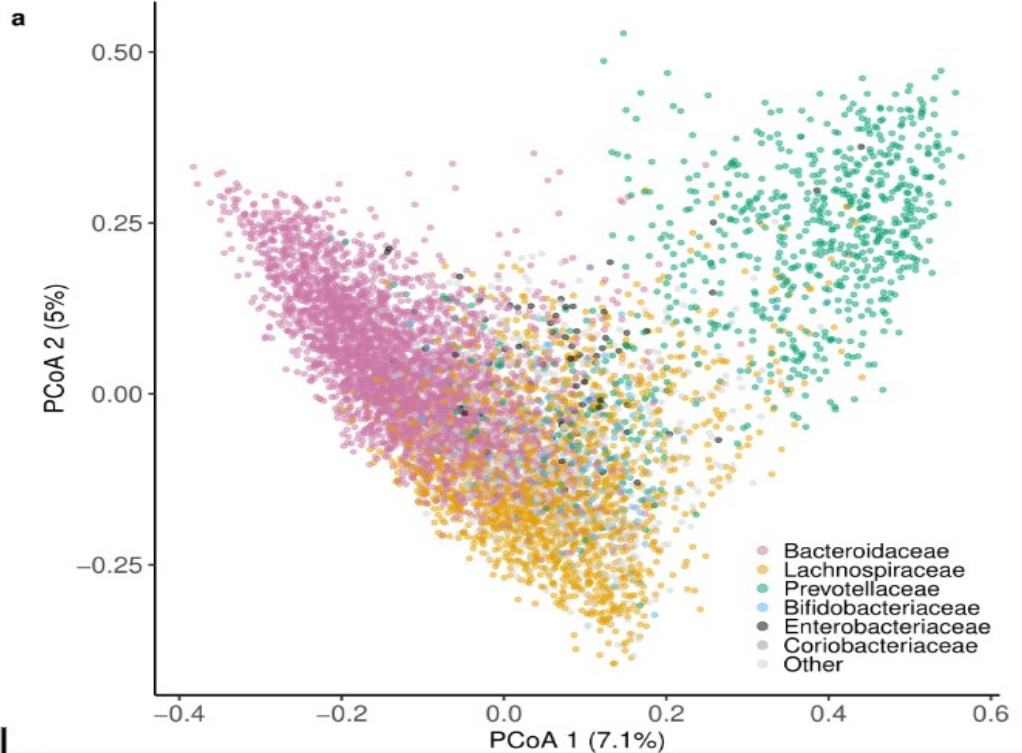
Software for the Integration of Multiomics Experiments in
Bioconductor **FREE**

Marcel Ramos; Lucas Schiffer; Angela Re; Rimsha Azhar; Azfar Basunia; Carmen Rodriguez; Tiffany Chan; Phil Chapman;
Sean R. Davis; [David Gomez-Cabrero](#); Aedin C. Culhane; Benjamin Halbe-Kains; Kasper D. Hansen; Hanish Kodali; Marie S. Louis;
Arvind S. Mer; Markus Riemer; Martin Morgan; Vince Carey; Levi Waldron

Supporting standardization, reproducibility

runMDS()

plotFeatureLoadings()



Basic data operations

Abundance data:

Feature information:

Sample information:

Alternative experiments:

Ordinations:

Subsetting:

Transformation:

```
assay(tse, "counts")
```

```
rowData(tse)
```

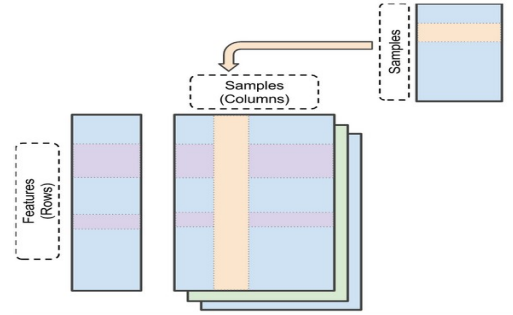
```
colData(tse)
```

```
altExp(tse, "Genus")
```

```
reducedDim(altExp(tse, "Genus"), "NMDS")
```

```
tse[1:5, 1:3]
```

```
transformAssay(tse, "clr", pseudocount=1)
```



Building modular workflows

Importing data

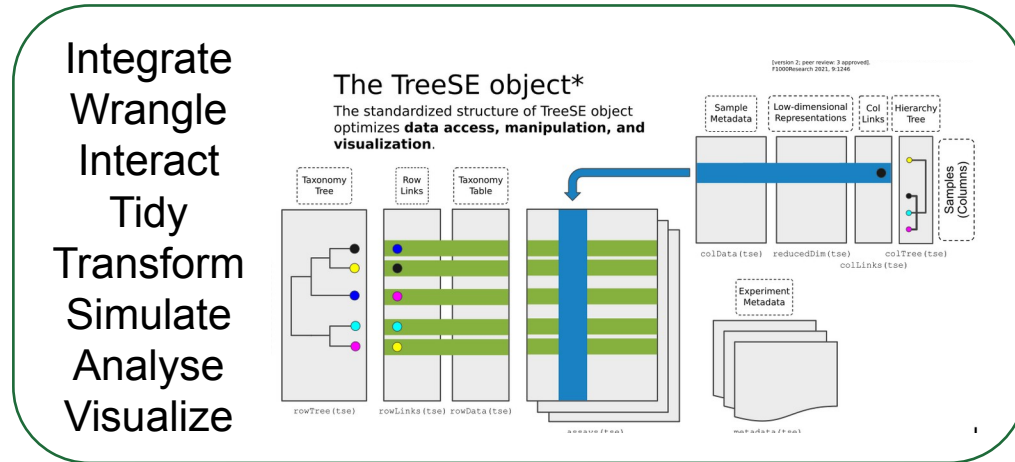
Standard formats



Open data

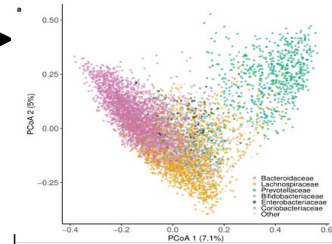
EBI/MGnify, HoloFood,
curatedMetagenomicData
ExperimentHub

Interoperable methods



Reproducible reporting

runMDS()

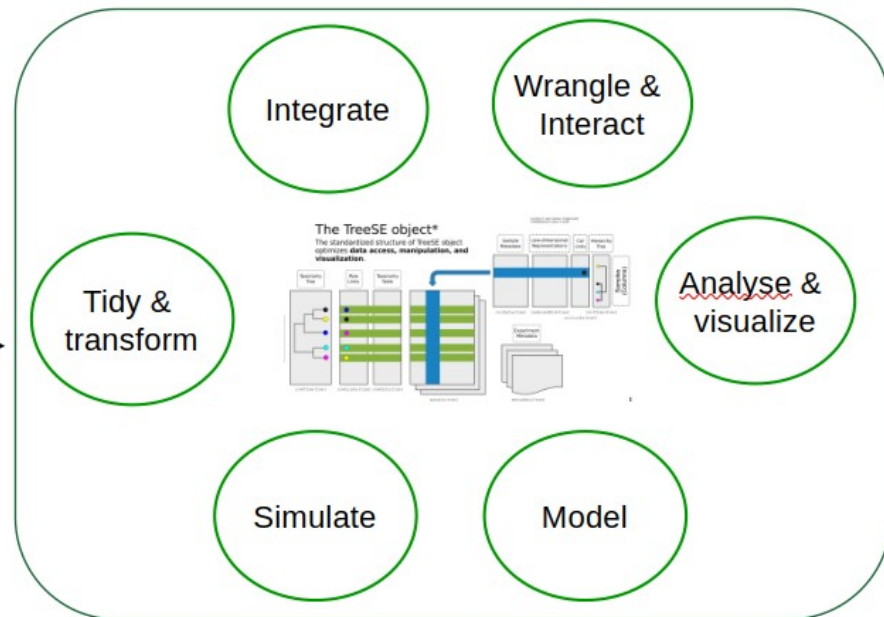


Importing data to standard data structures
facilitates collaborative methods development
& downstream analyses

Package ecosystem

Integration of taxonomic, metagenomic, functional, and multi-omic data

- ~20 packages suited for hologenome analysis
- many application domains use similar containers in R/Bioconductor: single cell, transcriptomics, proteomics, metabolomics



The tidyomics ecosystem: enhancing omic data analyses

[William J. Hutchison](#), [Timothy J. Keyes](#), [The tidyomics Consortium](#), [Helena L. Crowell](#), [Jacques Serizay](#), [Charlotte Soneson](#), [Eric S. Davis](#), [Noriaki Sato](#), [Lambda Moses](#), [Boyd Tarlinton](#), [Abdullah A. Nahid](#), [Miha Kosmac](#), [Quentin Clayssen](#), [Victor Yuan](#), [Wancen Mu](#), [Ji-Eun Park](#), [Izabela Mamede](#), [Min Hyung Ryu](#), [Pierre-Paul Axisa](#), [Paulina Paiz](#), [Chi-Lam Poon](#), [Ming Tang](#), [Raphael Gottardo](#), [Martin Morgan](#), ... [Stefano Mangiola](#) ✉ [+ Show authors](#)

[Nature Methods](#) 21, 1166–1170 (2024) | [Cite this article](#)

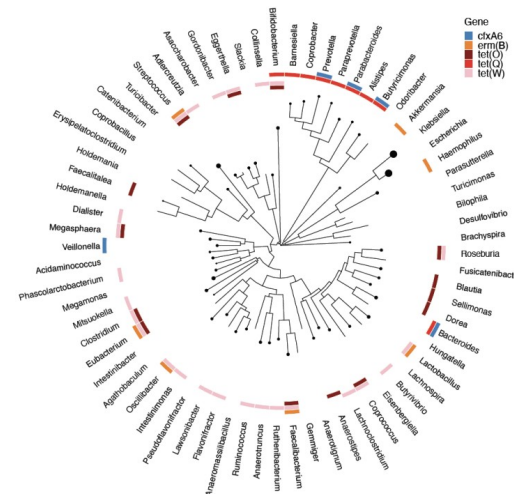


Orchestrating Microbiome Analysis



Online Gitbook (OMA) - beta version

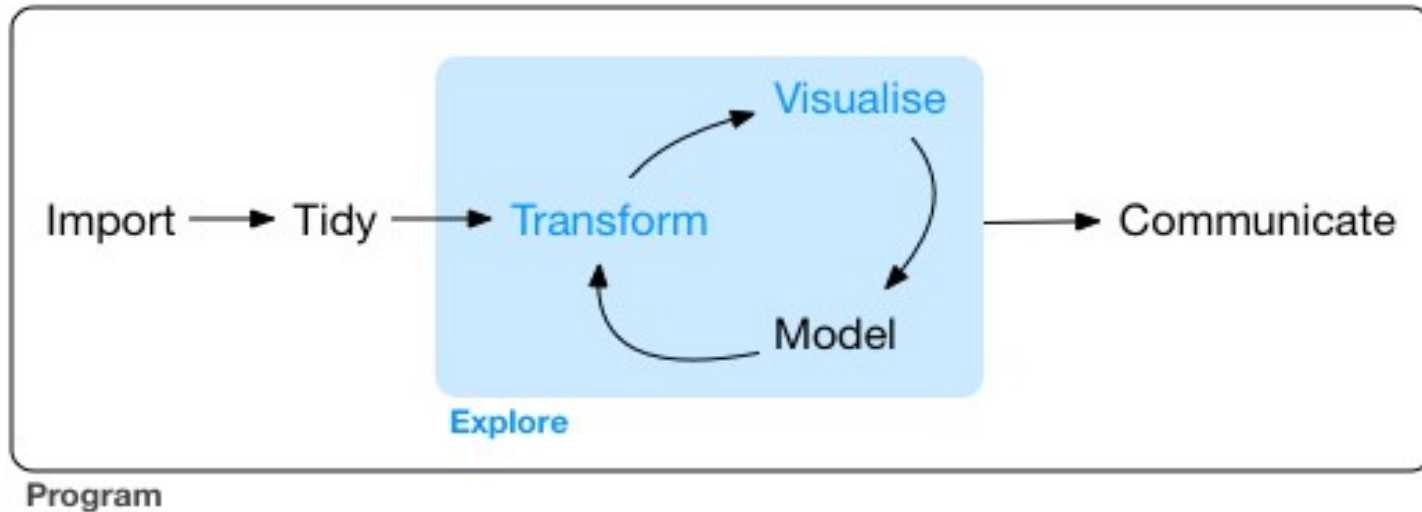
- 1) Downstream data analysis
- 2) Statistical programming workflows
- 3) Multi-assay methods



Learning and teaching biological data science in the Bioconductor community

Jenny Drnevich , Frederick J. Tan , Fabricio Almeida-Silva, Robert Castelo, Aedin C. Culhane, Sean Davis, Maria A. Doyle, Ludwig Geistlinger, Andrew R. Ghazi, Susan Holmes, Leo Lahti, Alexandru Mahmoud, Kozo Nishida, Marcel Ramos, Kevin Rue-Albrecht, David J. H. Shih, Laurent Gatto , Charlotte Soneson  [[view less](#)]

Evidence-based recommendations on best methods & practices



Good enough practices in scientific computing

Greg Wilson  , Jennifer Bryan , Karen Cranston , Justin Kitzes , Lex Nederbragt , Tracy K. Teal 

Published: June 22, 2017 • <https://doi.org/10.1371/journal.pcbi.1005510>

microbiome.github.io/OMA

